

Anusha Upadhyay

anushaupadhyay1111@gmail.com · +1 (856) 655 4024 · [linkedin.com/in/upadhyay-anusha/](https://www.linkedin.com/in/upadhyay-anusha/) · US Citizen · She/Her

MS CS candidate at NC State University who builds full-stack applications, data pipelines, and production ML systems, with experience running both online and offline evaluations. Hands-on experience across Python, Java, C++, and JavaScript/React, with PyTorch and TensorFlow for modeling, FAISS for retrieval, and SQL, databases, APIs, and cloud tooling.

EDUCATION

North Carolina State University

Expected Dec 2026

Master's (MCS) in Computer Science (Concentration in Data Science)

Software Engineering, Data Analysis, Data Structures, Neural Networks, Human-Computer Interaction, DevOps, Computer Graphics, Deep Learning, Database Management

Vellore Institute of Technology (VIT), Vellore

Sept 2021-May 2025

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming, Artificial Intelligence, Natural Language Processing

EXPERIENCE

Neuraluna AI

June 2026-August 2026

Software Engineering Intern (AI Evaluation Engineering)

- Built an automated LLM benchmarking harness evaluating 32 models across 8 AI agents on quality, cost, and latency to replace intuition-based model selection with data-driven picks.
- Applied Pareto frontier analysis to flag a dominated production model, recommending a swap that cut cost ~33% while improving quality and latency.
- Audited an internal AI agent-builder's prompt and config, validating fixes via A/B testing and shipping 6 confirmed improvements.

North Carolina State University

February 2026-Present

Research Assistant (Applied ML & Software Development)

- Conduct statistical analysis of large-scale social media engagement datasets to identify behavioral and interaction patterns.
- Develop and implement predictive modeling pipelines to analyze engagement trends and platform-level performance metrics.
- Support the design and evaluation of data-driven studies examining digital content and user interaction dynamics.
- Perform exploratory data analysis and build analytical workflows to translate quantitative findings into actionable insights.

Temple University, Philadelphia

January 2025- May 2025

Research Assistant

- Built a multimodal, multilingual RAG platform over 1,000+ medical documents to improve accessibility of colorectal cancer information.
- Designed the ingestion/embedding pipeline and implemented FAISS-based semantic search to boost retrieval precision.
- Fine-tuned and evaluated 5 LLM configurations, improving medical response quality and factual grounding.

Institute of Engineering and Technology

March 2022- January 2024

Web Developer & UX Designer at IET Vellore

- Designed UX and implemented front-end features for flagship event websites, improving usability and navigation.
 - Collaborated with design and technical teams to deliver functional event platforms within project timelines.
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ACADEMIC PROJECTS

Wearable Acoustic Event Classification (NCSU)

Jan 2026 – May 2026

- Built a 4-class CNN + LightGBM ensemble on PANNs audio embeddings to classify acoustic events from chest-mic recordings.
- Achieved 0.78 Macro F1 / 0.93 accuracy on held-out subjects; scored top 5 in a class competition.

Skin Cancer Detection with EfficientNet-B3 + ViT Ensemble (NCSU)

Jan 2026 – May 2026

- Built an EfficientNet-B3 + ViT ensemble (focal loss, MixUp/CutMix, test-time augmentation) to classify dermatoscopic skin lesion images.
- Improved Val AUC from 0.94 to 0.96, reaching 93.9% recall on malignant cases; presented at the department's student symposium.

ReqTrace – AI-Powered Requirements Intelligence Platform (NCSU)

Aug 2025 – Dec 2025

- Processed unstructured meeting recordings into structured data using Python and NLP pipelines.
 - Built a hybrid ML system combining semantic embedding/similarity search, vector search (FAISS), and graph analytics (Neo4j) for automated traceability and impact analysis.
 - Delivered an end-to-end ML-powered prototype, ranked Top 3 in the class.
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TECHNICAL SKILLS

- Programming Languages: C++, Java, Python, SQL, R, JavaScript, C
- Databases: MySQL, MongoDB, Google BigQuery
- Web Technologies: HTML, CSS, React, Streamlit
- Cloud & DevOps: Google Cloud Functions, Kubernetes, Vision AI, Cloud Armor
- Data Analytics & Visualization: Pandas, NumPy, Matplotlib, Tableau, Looker Studio
- Machine Learning & Generative AI: TensorFlow, Scikit-learn, NLP, LLMs, RAG, Vector Search, Ollama, FAISS